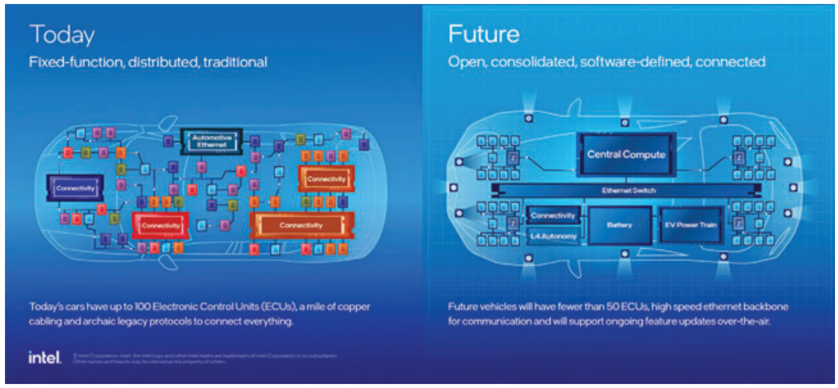
enable in-vehicle AI, such as GenAI and camera-based driver/passenger monitoring. The first generation of the Intel SDV SoC family will have up to 12 cores with a power envelope ranging from 12W to 45W. Intel showed its vision for the future of electric vehicles by comparing it to the current EV architecture and how it sees the overall architecture changing into something simpler, connected, centralized, and software-defined. You can draw a parallel to the PC architecture, which is focused on the central computing model. Intel sees the EV architecture with a similar holistic lens that revolves around the central computing unit and puts Intel right into the driver's seat.



Intel's AI 2024 lineup

Intel Meteorlake has kicked off Intel's AI journey to an impressive start, but the upcoming Arrow Lake and Lunar Lake series are expected to crank up the AI performance to the next level. The Arrow Lake products will bring high-performance AI to the gaming lineup, with the first desktop gaming CPU with an AI accelerator. It will be interesting to see what Intel has in store, especially for a gaming desktop CPU that boasts of AI acceleration. We're keeping a close eye on this one. On the notebook (or laptop) side, the Lunar Lake will be the one to watch out for. Intel claims it has been built on a "radically new low-power architecture". From what we know through the rumor mills and leaks, Lunar Lake will be among the first Intel chips to feature memory on package (MoP) design. That should help in lowering power consumption and bring along significant performance gains.

Intel shows off Meteorlake

At the workshop, Intel had demos of on-device AI workloads that showed the performance prowess of the Intel Core Ultra series of laptop processors. A longer chat with Intel engineers gave me an interesting insight into AI compute management and how the Intel thread director manages AI workloads. While the talk and focus have been on the dedicated NPU cores in the Intel Core Ultra Meteorlake series, the AI compute tasks are not always just managed by the NPU cores. The NPU cores are the low-powered always-listening type workhorses that will manage the basics, but when it comes to heavy lifting, the P-cores (Performance cores) and the brand-new Intel ARC GPU are the ones that spring into action. And it does make a lot of sense to push the right workloads to the right kind of cores and subsystems. Especially when it comes to extracting or delivering better performance across a wider range of productivity tasks and applications that are and will increasingly use generative AI to improvise and automate output.

Intel is ready to game!

Besides the AI workloads, we also checked out the gaming performance of the Intel Core Ultra-based laptops. The Intel ARC GPU performance combined with Intel XeSS super sampling optimization dished out some impressive frame rate gains over the previous generation (13th Gen) Intel processors. And that is good news for gamers; you can now expect to fire some rounds of your favorite FPS games on that Intel Core Ultra-based laptop without worrying about frame rates. The Intel Core Ultra based laptops using the Intel ARC IGP show about 2x performance jump when it comes to real-world gaming and can deliver up to 3x more frames when using XeSS. However, we do think that the XeSS performance would depend on specific game titles that are optimized to take full advantage of Intel XeSS. That said, the big story here is that we are not talking about the usual 20-30% performance jump over the previous generation. This is a 2x/3x performance gain and that too in the graphics performance department, which has traditionally not been an Intel stronghold.

Intel also announced its hand-held gaming console play at CES 2024 with the MSI Claw A1M, a Windows-based hand-held gaming console powered by an Intel Core Ultra processor. We spent a few minutes gaming on the MSI Claw A1M and liked what we saw; the games were loading quickly, and the performance was excellent at 1080p. The device itself felt slightly better built in comparison to the Asus ROG Ally, which went on sale in India in 2023 and is powered by the AMD Ryzen platform. It will be interesting to see if MSI does bring the Claw to India and how it is priced in comparison to the competition. d